

Tissue	GSE	Samples	Genes	Classes	Shannon entropy
Lung	GSE60052	86 [79 tumor, 7 normal]	58735	2	0.407

GSE60052 - With the 10 Top Genes & Hyperparameter Optimization						
Classifier	Accuracy	F1-score	Lr+	Lr-	Sensitivity	Specificity
DT	0.983±0.006	0.991±0.004	6.949±0.046	0.007±0.008	0.994±0.007	0.857±0.000
SVM	0.970±0.007	0.983±0.003	4.601±1.655	0.017±0.002	0.987±0.000	0.769±0.080
NB	0.977±0.000	0.987±0.000	6.902±0.000	0.015±0.000	0.987±0.000	0.857±0.000
KNN	0.977±0.000	0.987±0.000	6.902±0.000	0.015±0.000	0.987±0.000	0.857±0.000
LR	0.965±0.000	0.981±0.000	3.451±0.000	0.018±0.000	0.987±0.000	0.714±0.000
MLP	0.922±0.015	0.958±0.008	1.437±0.221	0.083±0.065	0.977±0.016	0.304±0.109
RF	0.974±0.006	0.986±0.003	6.883±0.042	0.018±0.007	0.984±0.006	0.857±0.000
DUM	0.919±0.000	0.958±0.000	1.000±0.000	0.000±0.000	1.000±0.000	0.000±0.000

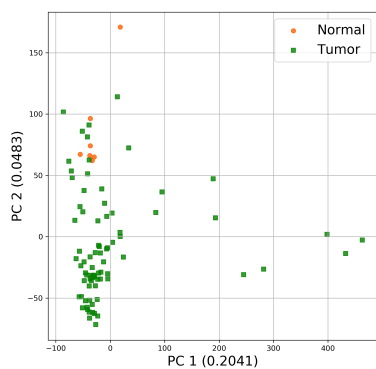
GSE60052 - With all Genes & Hyperparameter Optimization						
Classifier	Accuracy	F1-score	Lr+	Lr-	Sensitivity	Specificity
DT	0.965±0.012	0.981±0.006	6.813±0.087	0.030±0.014	0.974±0.012	0.857±0.000
SVM	0.954±0.004	0.976±0.002	2.304±0.009	0.021±0.007	0.988±0.004	0.571±0.000
NB	0.919±0.000	0.958±0.000	1.000±0.000	0.000±0.000	1.000±0.000	0.000±0.000
KNN	0.942±0.000	0.968±0.000	6.636±0.000	0.060±0.000	0.949±0.000	0.857±0.000
LR	0.953±0.000	0.975±0.000	2.301±0.000	0.023±0.000	0.987±0.000	0.571±0.000
MLP	0.894±0.019	0.944±0.011	1.026±0.104	0.232±0.086	0.969±0.022	0.048±0.088
RF	0.940±0.010	0.968±0.005	1.661±0.322	0.029±0.021	0.990±0.006	0.383±0.119
DUM	0.919±0.000	0.958±0.000	1.000±0.000	0.000±0.000	1.000±0.000	0.000±0.000

GSE60052 - With all Genes & without Hyperparameter Optimization						
Classifier	Accuracy	F1-score	Lr+	Lr-	Sensitivity	Specificity
DT	0.967±0.011	0.982±0.006	6.830±0.079	0.027±0.013	0.977±0.011	0.857±0.000
SVM	0.919±0.000	0.958±0.000	1.000±0.000	0.000±0.000	1.000±0.000	0.000±0.000
NB	0.919±0.000	0.958±0.000	1.000±0.000	0.000±0.000	1.000±0.000	0.000±0.000
KNN	0.942±0.000	0.968±0.000	6.636±0.000	0.060±0.000	0.949±0.000	0.857±0.000
LR	0.953±0.000	0.975±0.000	2.301±0.000	0.023±0.000	0.987±0.000	0.571±0.000
MLP	0.881±0.024	0.936±0.013	1.014±0.102	0.306±0.208	0.955±0.024	0.051±0.079
RF	0.947±0.009	0.972±0.005	1.950±0.487	0.025±0.011	0.989±0.004	0.470±0.099
DUM	0.919±0.000	0.958±0.000	1.000±0.000	0.000±0.000	1.000±0.000	0.000±0.000

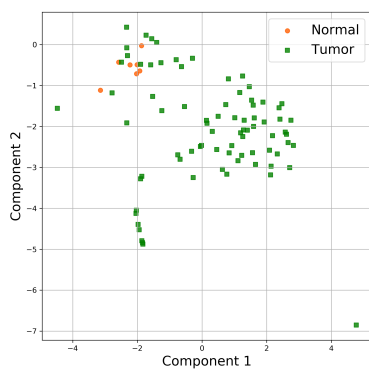
GSE60052 - With the 10 Top Genes & Hyperparameter Optimization & Class weight						
Classifier	Accuracy	F1-score	Lr+	Lr-	Sensitivity	Specificity
DT	0.988±0.000	0.994±0.000	6.993±0.000	0.000±0.000	1.000±0.000	0.857±0.000
SVM	0.969±0.008	0.983±0.004	5.109±1.749	0.020±0.008	0.985±0.006	0.783±0.073
LR	0.988±0.000	0.994±0.000	0.000±0.000	0.013±0.000	0.987±0.000	1.000±0.000
RF	0.965±0.007	0.981±0.003	6.499±1.029	0.028±0.007	0.977±0.005	0.843±0.043

GSE60052 - With the 10 Top Genes & Hyperparameter Optimization & SMOTE						
Classifier	Accuracy	F1-score	Lr+	Lr-	Sensitivity	Specificity
DT	0.998±0.003	0.998±0.003	0.000±0.000	0.005±0.006	0.995±0.006	1.000±0.000
SVM	0.988±0.003	0.988±0.003	0.000±0.000	0.024±0.006	0.976±0.006	1.000±0.000
NB	0.994±0.000	0.994±0.000	0.000±0.000	0.013±0.000	0.987±0.000	1.000±0.000
KNN	0.987±0.000	0.987±0.000	0.000±0.000	0.025±0.000	0.975±0.000	1.000±0.000
LR	0.994±0.000	0.994±0.000	0.000±0.000	0.013±0.000	0.987±0.000	1.000±0.000
MLP	0.962±0.011	0.961±0.011	57.068±22.437	0.063±0.019	0.938±0.018	0.985±0.016
RF	0.989±0.003	0.989±0.003	75.923±0.000	0.020±0.006	0.980±0.006	0.999±0.003
DUM	0.000±0.000	0.000±0.000	0.000±0.000	0.000±0.000	0.000±0.000	0.000±0.000

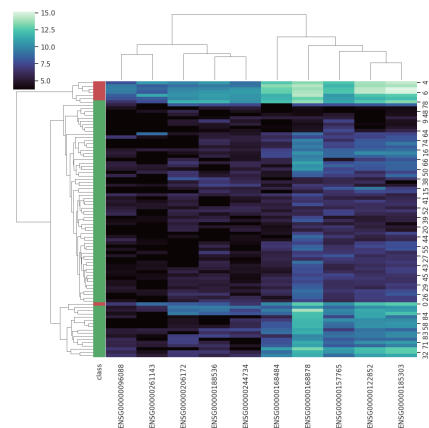
<https://sbc.b.inf.ufrgs.br/barracurda>



(a) PCA GSE60052



(b) t-SNE GSE60052



(c) HeatMap GSE60052 - Top 10 Genes

<https://sbc.b.inf.ufpr.br/barracurda>